

Composition Pair B1.09 + B1.14

The Sequentially Co-Required Pair Where Birth Creates the Governance-Anchored Entity With Initial DNA as the Starting Point for Directed Selection, and Directed Selection Requires a Properly-Born Entity's Governed DNA as Its Modification Substrate

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Date: May 13, 2026

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026).

Abstract

This note formalizes the composition relationship between B1.09 (birth — governed origination) and B1.14 (directed selection — human-governed DNA modification) as a sequentially co-required pair in which B1.09 is architecturally prior. The pair exhibits asymmetric sequential dependency: B1.09 must precede B1.14 for a given entity because birth creates the initial DNA specification, the governance provenance anchor, and the lineage chain starting point that directed selection subsequently extends. Without birth, directed selection modifies DNA whose provenance history has no governed origin. The pair also exhibits mutual enabling in the forward direction: without B1.14, the entity born through B1.09 cannot improve its governing specifications through governance, and its initial DNA becomes static over operational time. The DNA version chain per B2.69 is identified as the structural embodiment of the pair — version 1 created at birth, versions 2 through N created through directed selection events, the complete chain encoding both commitments simultaneously. A practical quality relationship follows: investment in birth specification quality per B2.40 reduces the directed selection burden required to achieve operational effectiveness, making B1.09 execution quality a predictor of B1.14 effort. Failure modes for each absence direction are named: Evolution Stasis (B3.24) when B1.14 is absent, Ungoverned Birth (B3.10) when B1.09 is absent, and Ungoverned DNA Modification (B3.15) when B1.14's governance process is bypassed.

1. Pair Identification

The composition pair formalized in this note is:

B1.09 — Birth: governed origination establishing governance provenance, initial DNA specification per B2.40, and lineage anchor per B2.43.

B1.14 — Directed selection: human-governed DNA modification evolving the entity's governing specifications after birth.

Relationship type: Sequentially co-required pair, with B1.09 architecturally prior.

This pair differs from mutually enabling pairs elsewhere in Series B4 in that the dependency is asymmetric in temporal direction. B1.09 must precede B1.14 for a given entity — not merely that both must eventually be present, but that birth is the necessary prior condition for any directed selection over the same entity's DNA. The pair is not interchangeable: B1.14 cannot occur first and B1.09 second. The architectural prior is fixed by the logic of governed origination: an entity that does not yet exist cannot have its DNA modified.

2. Enabling Direction: B1.09 Is Architecturally Prior

Birth per B1.09 creates three structures that directed selection per B1.14 subsequently requires.

Initial DNA per B2.40. The birth event authors the entity's initial specification — the orchestration rules, behavioral parameters, and governing content that define how the entity will operate at the moment it begins its operational life. This specification is DNA version 1 in the version chain per B2.69. It is not a placeholder; it is the best-available governance knowledge at birth time, authored under the authority architecture the birth event invokes. Every directed selection event that follows modifies this initial specification or its successors. Version 1 must exist before version 2 is possible.

Governance provenance per A2.40. The birth record created through B1.09 establishes the governance provenance chain for the entity. Every subsequent governance event — including every directed selection event — extends this chain. The chain documents who authorized each change, under what authority, and at what time. A provenance chain has exactly one beginning; that beginning is the birth record. Without it, the chain has no governed anchor, and the DNA authority claim per A1.08 — that the substrate is the source of truth — is weakened at its root.

Lineage anchor per B2.43. Birth creates the lineage anchor that places the entity in its governance lineage. Directed selection events extend the lineage chain from this anchor. Where an entity has parents produced through mating, the birth record records those parents; where it is an original creation, the birth record records the governance act of origination. Either way, the lineage chain starts at the birth record. Directed selection events add to the chain; they do not create it.

These three structures — initial DNA, governance provenance, lineage anchor — are co-created in the birth event and are all required by directed selection to function with full governance integrity. This is

why B1.09 is architecturally prior rather than merely conventionally prior: the structures that directed selection operates on do not exist until birth creates them.

3. Mutual Dependency: Why Each Requires the Other

B1.09 requires B1.14 for entity evolution. An entity born through B1.09 has initial DNA that represents the best-available governance knowledge at birth time. But governance knowledge improves over operational time. Requirements change. The entity's operating environment evolves. Gaps in the initial specification become visible only through operational experience. Without B1.14, the entity's DNA is static after birth — frozen at the quality of the birth specification regardless of how much better governance understanding subsequently becomes. Birth per B1.09 initializes the governed entity; directed selection per B1.14 is the mechanism through which the initialized entity continues to improve. Together, the pair describes the complete governed lifecycle from initialization through ongoing evolution.

B1.14 requires B1.09 for provenance-grounded substrate. Directed selection per B1.14 modifies DNA by creating new versions that extend the version chain per B2.69. Each new version carries governance records per A2.40. But the chain must start at a governed point. If the entity was not properly born per B1.09 — if its origination bypassed governed birth — the DNA version chain has no governed beginning. Directed selection then adds governed links to a chain whose starting point is unanchored. The DNA authority claim per A1.08, which requires that every piece of authoritative state be traceable through the substrate, is weakest precisely at the foundation: the initial specification whose provenance is missing.

This mutual dependency is why the pair is co-required rather than merely sequentially ordered. Sequential ordering alone would say: first B1.09, then B1.14. Co-requirement adds: the absence of either, even after the proper sequence begins, degrades the architecture's integrity. Birth without subsequent directed selection produces an entity that cannot improve. Directed selection applied to an entity that was not properly born produces an improvement process without a governed starting point.

4. The DNA Version Chain as Pair Embodiment

The DNA version chain per B2.69 is the structural artifact in which this pair's co-requirement is concretely embodied.

Version 1 is created at birth per B1.09. It contains the initial specification per B2.40, the governance provenance from the birth record per A2.40, and the lineage anchor per B2.43. It is the governed starting point: the first link in the chain, authored under the authority the birth event invokes.

Versions 2 through N are each created through a directed selection event per B1.14. Each version extends the provenance chain from version 1, adding an authorization record per A2.47, an authoring record per A2.04, and a change record per A2.40. Each version is traceable back to version 1 through the unbroken chain. Each represents the entity's governing specifications at the moment of a

governance-authorized improvement.

The complete chain — from version 1 through the current version — embodies both commitments simultaneously. Remove version 1 and the chain has no governed beginning; the B1.09 commitment is missing. Remove all versions after version 1 and the entity cannot evolve; the B1.14 commitment is missing. An intact chain, growing over the entity's operational life through authorized directed selection events, is the observable signature that both commitments are present and operating correctly.

This makes the version chain not merely a useful operational artifact but a pair-integrity diagnostic. Inspection of the version chain answers three governance questions: Does version 1 carry birth provenance? Do subsequent versions each carry directed selection provenance? Is the chain unbroken from version 1 to the current version? Affirmative answers to all three indicate both pair members are functioning. Negative answers identify which member is absent or impaired.

5. Birth Specification Quality and Directed Selection Effort

The pair exhibits a quality relationship with practical operational implications. The quality of the initial DNA specification per B2.40 — the B1.09 output — directly affects the directed selection effort per B1.14 required to achieve operational effectiveness.

A high-quality birth specification accurately captures governance intent for the entity at the time of birth. It specifies orchestration rules precisely enough that the entity's early operation closely matches governance expectations. The gap between initial specification and operational optimum is small. Directed selection events are corrective at the margins — refinements and incremental improvements rather than fundamental rework.

A low-quality birth specification either misses governance intent, specifies rules imprecisely, or leaves significant orchestration gaps. The entity operates with specifications that diverge from governance expectations from the outset. Directed selection events must first correct foundational misspecifications before they can improve on the initial design. The directed selection burden is higher because the starting point is farther from the operational optimum.

This quality relationship operates in one direction only: B1.09 quality affects B1.14 burden. The reverse does not hold — directed selection quality does not affect what birth would have needed to specify. This asymmetry is another expression of the architectural prior: B1.09 sets the starting conditions; B1.14 operates from those starting conditions forward.

The practical implication is that governance investment in birth specification quality pays compound returns. Better initial specifications reduce directed selection effort at every subsequent version, not just the first correction cycle. The investment is made once at birth; the benefit accrues across the entity's entire operational life. Governance that treats birth specification as the first act of an ongoing improvement process — rather than as a one-time initialization to be corrected later — reduces total lifecycle governance cost.

6. Co-Presence Requirement

B1.09 present, B1.14 absent — birth without directed selection. The entity is properly born with complete initial DNA, governance provenance, and lineage anchor. But without directed selection, its DNA cannot improve through governance. Operational experience accumulates; the entity's governing specifications do not incorporate it. The initial specification ages relative to the entity's actual operating environment. Over time, the gap between current specifications and governance intent grows without a mechanism for correction. This is the Evolution Stasis failure mode per B3.24 at entity scope: the entity exists and operates under governance, but its specifications cannot improve.

B1.14 present, B1.09 absent — directed selection without proper birth. Directed selection modifies DNA for an entity whose origination bypassed the birth governance process. The modification creates new DNA versions and adds governance records for those versions. But the version chain begins at an unanchored point: there is no governed version 1 from which the chain traces. The DNA authority claim per A1.08 is weakened because the substrate cannot trace authoritative state to its governed origin. Operational integrity may be maintained in practice, but the architecture's provenance guarantee is incomplete. This is the Ungoverned Birth pattern per B3.10 followed by directed selection that cannot retroactively supply the missing provenance anchor.

Full co-presence — birth followed by directed selection. The entity is properly born per B1.09 with complete initial DNA and governance provenance. Directed selection per B1.14 creates subsequent versions extending the version chain with authorized improvements. The entity's governing specifications improve over its operational life under governance. The DNA authority claim per A1.08 is complete: every version is traceable to the governed beginning. This is the complete governed entity lifecycle — from initialization through evolution — that the pair describes when both commitments are present.

7. Operational Integration

The pair operates as a sequential governance workflow across the entity's lifecycle:

Governance decision to create entity → birth per B1.09 → initial specification per B2.40 authored → birth record per A2.40 created → DNA version 1 established per B2.69 → entity enters operation → operational experience accumulates → governance identifies improvement opportunity → directed selection event per B1.14 initiated → authorization per A2.47 obtained → updated specification authored per A2.04 → change recorded per A2.40 → DNA version N created → entity operates under improved specifications → cycle continues for the entity's operational life.

The workflow is explicitly cyclic after the first directed selection event. Birth is a one-time event; directed selection recurs as governance knowledge improves and operational requirements evolve. Each cycle extends the version chain by one governed link. The total chain length at any point in time records the entity's complete governance history from origination through its current specifications.

8. Detection

Three verification points confirm the pair is present and operating:

B2.44 — birth verification. Confirms the entity's birth specification is complete: initial DNA present per B2.40, governance provenance present per A2.40, lineage anchor present per B2.43, version 1 established in the version chain per B2.69.

B2.72 — directed selection verification. Confirms that all DNA modifications after version 1 are authorized and recorded: each subsequent version carries an authorization record per A2.47, authoring record per A2.04, and change record per A2.40 traceable to version 1.

DNA version chain integrity per B2.69. Confirms the complete chain: version 1 carries birth provenance; each subsequent version carries directed selection provenance; the chain is unbroken from version 1 to the current version. This chain-level check is the pair-level diagnostic: integrity of the complete chain confirms both pair members have been functioning correctly across the entity's entire governed lifecycle.

9. Failure Modes

Ungoverned Birth (B3.10) — B1.09 is absent. The entity's origination bypasses the birth governance process. DNA version 1 lacks governance provenance. Directed selection may subsequently occur, but cannot retroactively supply the governed origin that birth per B1.09 creates. The version chain has no anchored beginning.

Evolution Stasis (B3.24) — B1.14 is absent. The entity is properly born but has no mechanism for improving its governing specifications through governance. The birth specification becomes the permanent specification regardless of how much operational experience or governance knowledge accumulates. The version chain has a governed version 1 and nothing after it.

Ungoverned DNA Modification (B3.15) — B1.14's governance process is bypassed. The entity was properly born and has an anchored version chain, but subsequent DNA modifications are made outside the governance architecture — without authorization per A2.47, without authoring records per A2.04, or without change records per A2.40. The version chain breaks: subsequent versions lack directed selection provenance. The DNA authority claim per A1.08 is weakened at the modification point even though the chain's origin is sound.

10. Conclusion

The B1.09 + B1.14 composition pair is the sequentially co-required pair through which the complete governed entity lifecycle is expressed. Birth per B1.09 initializes the entity by creating its initial DNA specification, establishing its governance provenance anchor, and anchoring its lineage chain. Directed selection per B1.14 evolves the entity by extending the DNA version chain through governance-authorized improvements over the entity's operational life. Neither commitment alone is

sufficient: birth without directed selection produces static specifications that age against a changing operational environment; directed selection without proper birth produces an improvement process without a governed foundation from which the version chain can trace.

The DNA version chain per B2.69 is the artifact in which this co-requirement is concretely visible: version 1 records the B1.09 commitment; versions 2 through N record the B1.14 commitment; the complete chain records both simultaneously. Birth specification quality per B2.40 sets the starting conditions for all subsequent directed selection effort, creating a practical quality relationship in which governance investment at birth time reduces the B1.14 burden across the entity's entire operational life.

Source Papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

How to Cite This Note

Li, W. (2026). *Composition Pair B1.09 + B1.14 — The Sequentially Co-Required Pair Where Birth Creates the Governance-Anchored Entity With Initial DNA as the Starting Point for Directed Selection, and Directed Selection Requires a Properly-Born Entity's Governed DNA as Its Modification Substrate*. May 13, 2026. ORCID: 0009-0004-8065-3235.